EXP 8: Implementing CloudFront for Caching and Application Security

Theory:

Amazon CloudFront is a content delivery network (CDN) operated by Amazon Web Services. The content delivery network was created to provide a globally distributed network of proxy servers to cache content, such as web videos or other bulky media, more locally to consumers, to improve access speed for downloading the content.

Steps to execute:

Step 1: Create an S3 Bucket

- Go to AWS Management Console.
- Open **Amazon S3**.
- Click **Create Bucket**.
- Enter a **bucket name**.
- Keep all other settings as default.
- Click **Create bucket**.

Step 2: Enable Static Website Hosting

- Open the created bucket.
- Go to the **Properties** tab.
- Scroll to **Static Website Hosting**.
- Click **Edit** → Enable it.
- Specify:
- Index document (e.g., index.html)
- Click **Save changes**.

Step 3: Upload Website Files

- Go to the **Objects** tab.
- Click **Upload**.
- Add your HTML file (e.g., index.html).
- Click **Upload**.

Step 4: Access S3 Website URL

- Go back to **Properties → Static Website Hosting**.
- Copy the **Bucket Website Endpoint URL**.
- Open it in a browser to verify the website.

Step 5: Create CloudFront Distribution

- Go to **Amazon CloudFront** service.
- Click **Create Distribution**.
- Set:
- Origin domain → Select your S3 bucket
- Create a **new OAI (Origin Access Identity)**.
- Keep all other settings as default.

- Click **Create Distribution**.

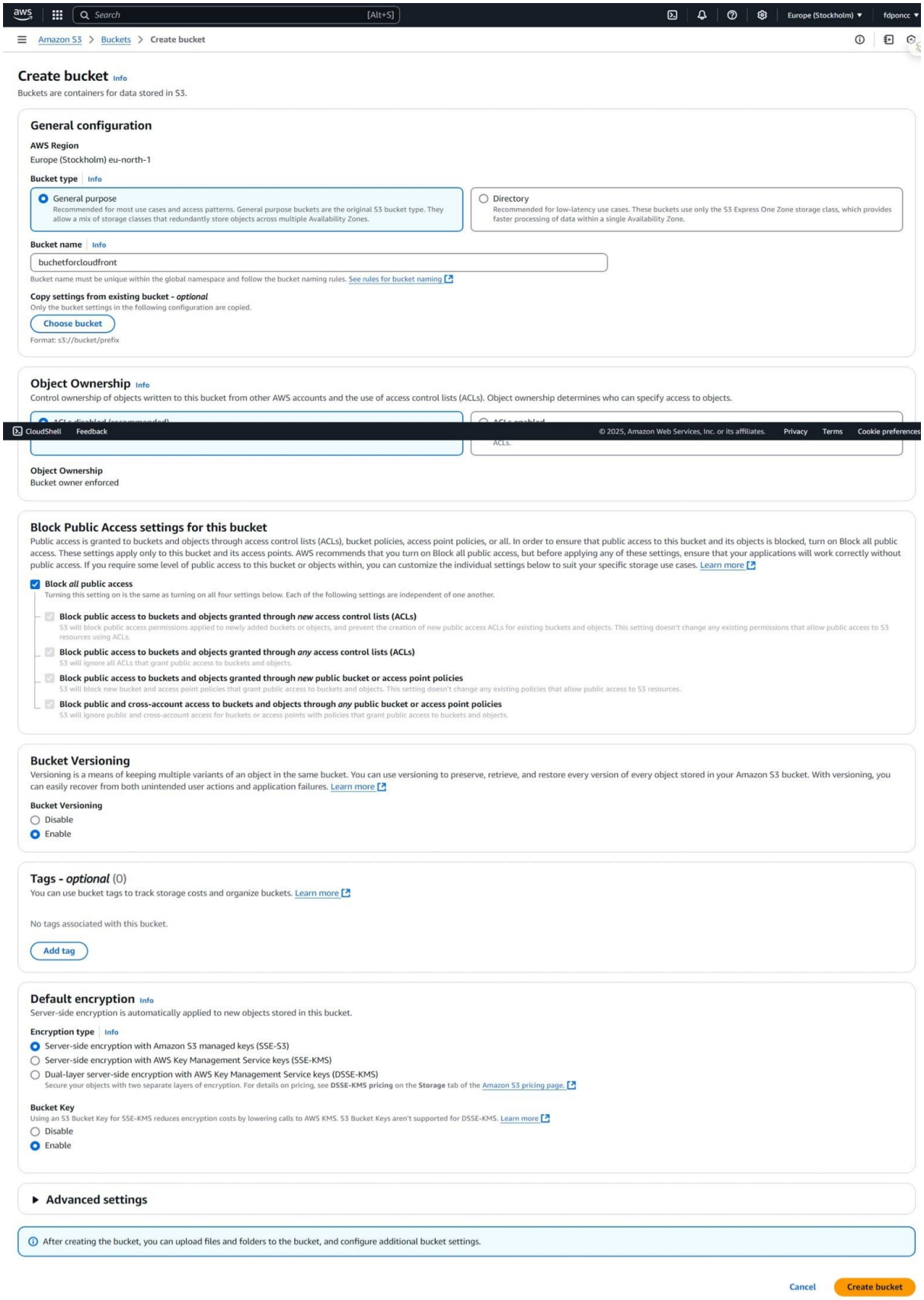
Step 6: Update Bucket Policy

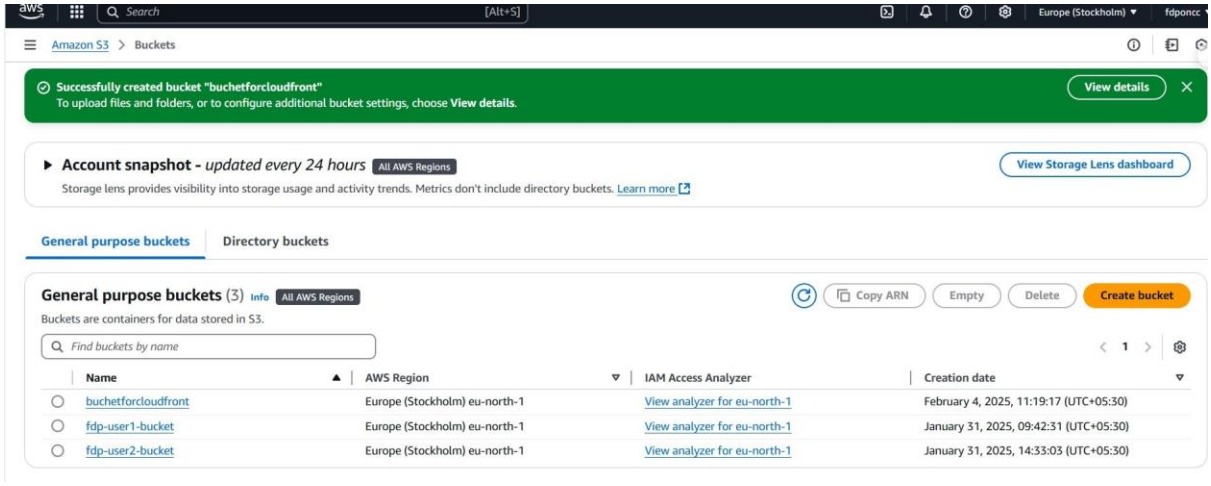
- Allow CloudFront access to the S3 bucket (policy created automatically when using OAI).

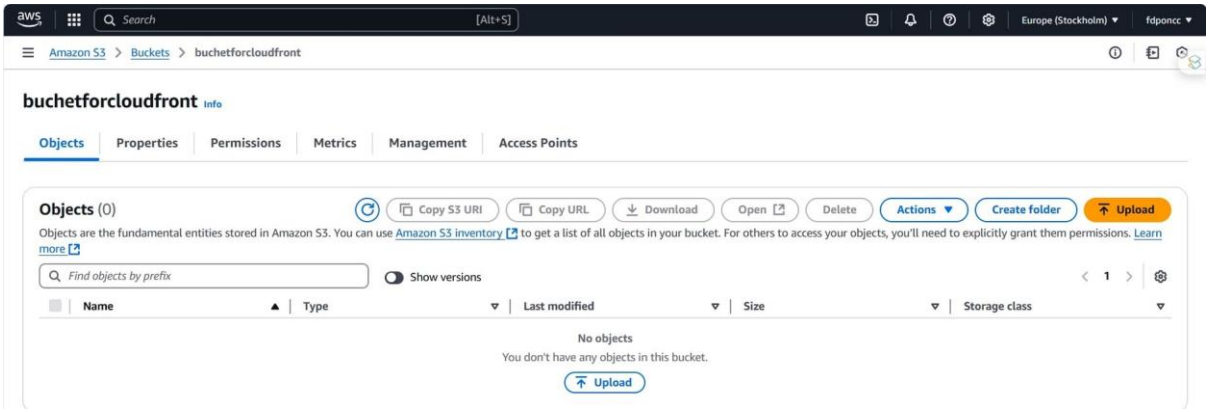
Step 7: Access via CloudFront

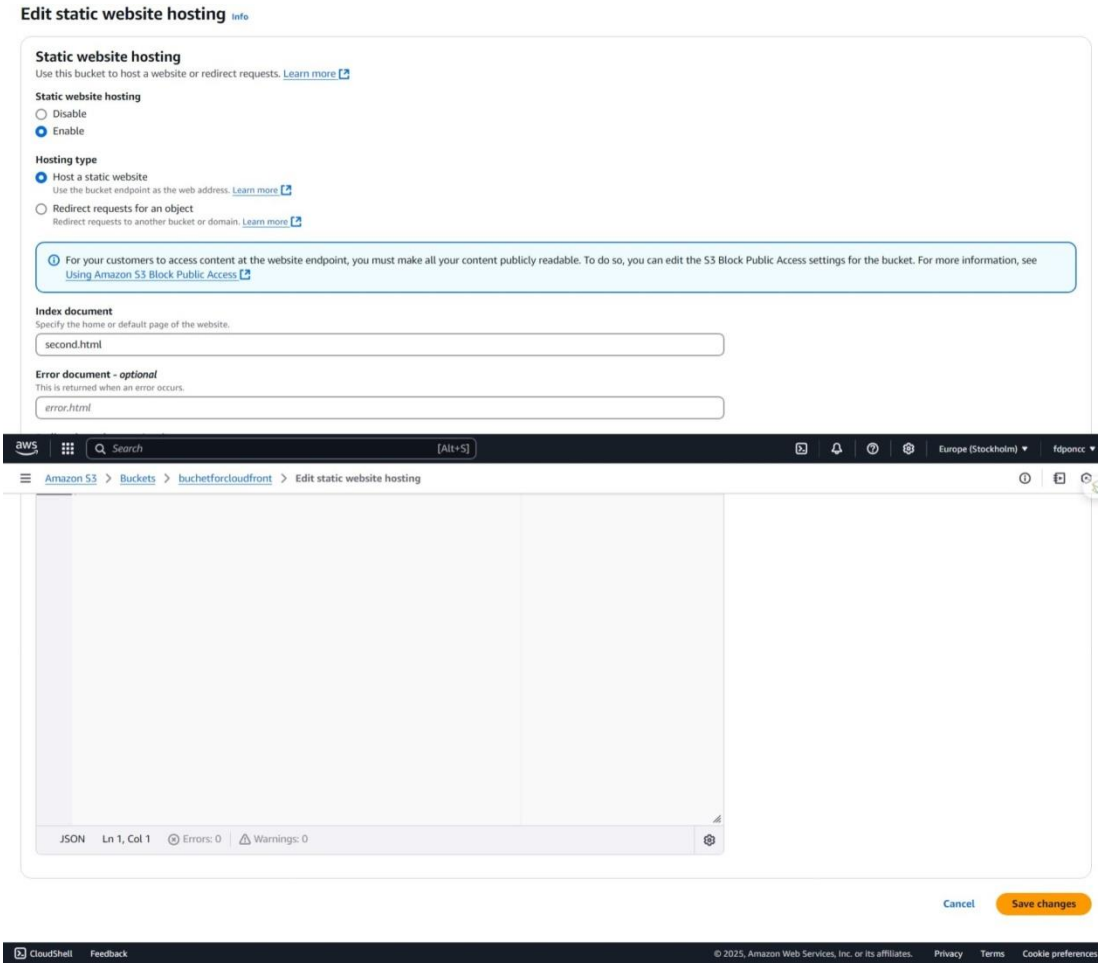
- After deployment, copy the **Distribution Domain Name**.
- Open it in a browser.
- Verify that the website loads via CloudFront (cached content).

EXP 8: Implementing CloudFront for Caching and Application Security




Amazon S3 > Buckets
Successfully created bucket "buchetforcloudfront"
To upload files and folders, or to configure additional bucket settings, choose View details.
View details
Account snapshot - updated every 24 hours
All AWS Regions
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. Learn more
View Storage Lens dashboard
General purpose buckets
Directory buckets
General purpose buckets (3) Info
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
Name
AWS Region
IAM Access Analyzer
Creation date
buchetforcloudfront
Europe (Stockholm) eu-north-1
View analyzer for eu-north-1
February 4, 2025, 11:19:17 (UTC+05:30)
fdp-user1-bucket
January 31, 2025, 09:42:31 (UTC+05:30)
fdp-user2-bucket
January 31, 2025, 14:33:03 (UTC+05:30)


Amazon S3 > Buckets > buchetforcloudfront
buchetforcloudfront Info
Objects
Properties
Permissions
Metrics
Management
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload


Edit static website hosting Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
second.html
Error document - optional
This is returned when an error occurs.
error.html
Amazon S3 > Buckets > buchetforcloudfront > Edit static website hosting
JSON
Ln 1, Col 1
Errors: 0
Warnings: 0
Cancel
Save changes
CloudShell
Feedback
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Privacy
Terms
Cookie preferences

aws | Search [Alt+S] | Europe (Stockholm) | fdponcc

Amazon S3 > Buckets > buchetforcloudfront > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 296.0 B)

Remove | Add files | Add folder

All files and folders in this table will be uploaded.

Find by name

Name	Folder	Type	Size
second.html	-	text/html	296.0 B

Destination Info

Destination

s3://buchetforcloudfront

▶ **Destination details**

Bucket settings that impact new objects stored in the specified destination.

▶ **Permissions**

Grant public access and access to other AWS accounts.

CloudShell | Feedback | © 2025, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences

▼ **Properties**

Specify storage class, encryption settings, tags, and more.

Storage class Info

Amazon S3 offers a range of storage classes designed for different use cases. Learn more or see Amazon S3 pricing

	Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
◉	Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
○	Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects >= 128 KB	-
○	Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
○	One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	General purpose or directory	1	30 days	128 KB	-	Per-GB fees apply
○	Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	General purpose	≥ 3	90 days	128 KB	-	Per-GB fees apply
○	Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	General purpose	≥ 3	90 days	-	-	Per-GB fees apply
○	Glacier Deep Archive	Long-lived archive data accessed less than once a year with retrieval of hours	General purpose	≥ 3	180 days	-	-	Per-GB fees apply
○	Reduced redundancy	Noncritical, frequently accessed data with milliseconds access (not recommended as S3 Standard is more cost effective)	General purpose	≥ 3	-	-	-	Per-GB fees apply

Server-side encryption Info

Server-side encryption protects data at rest.

Server-side encryption

◉ Don't specify an encryption key
The bucket settings for default encryption are used to encrypt objects when storing them in Amazon S3.

○ Specify an encryption key
The specified encryption key is used to encrypt objects before storing them in Amazon S3.

⚠ If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you upload objects. Otherwise, uploads will fail.

Checksums

Checksums are used for data integrity verification of new objects. Learn more

Checksum function

Checksum functions are used to calculate the checksum value. For objects smaller than 16 MB, only the full object checksum type is supported, for all checksum algorithms.

CRC64NVME (recommended)

Precalculated value - *optional*

When you provide a precalculated value for a single object, S3 compares it to the value it calculates using the selected checksum function. If the values don't match, the upload will fail. Learn more

Enter value

The precalculated value must be a Base64 encoded string. It must not exceed 128 characters, and can contain only letters (a-z, A-Z), numbers (0-9), forward slash (/), plus (+), or equals (=).

Tags - *optional*

You can use object tags to analyze, manage, and specify permissions for objects. Learn more

No tags associated with this resource.

Add tag

Metadata - *optional*

Metadata is optional information provided as a name-value (key-value) pair. Learn more

No metadata associated with this resource.

Add metadata

Upload succeeded
For more information, see the **Files and folders** table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination	Succeeded	Failed
s3://buchetforcloudfront	1 file, 296.0 B (100.00%)	0 files, 0 B (0%)

Files and folders | Configuration

Files and folders (1 total, 296.0 B)

Name	Folder	Type	Size	Status	Error
second.html	-	text/html	296.0 B	Succeeded	-

Amazon S3 > Buckets > buchetforcloudfront

Amazon S3

- General purpose buckets
- Directory buckets
- Table buckets
- Access Grants
- Access Points
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

- Dashboards
- Storage Lens groups
- AWS Organizations settings

Feature spotlight 10

Requester pays

Edit

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more

Requester pays
Disabled

Static website hosting

Edit

Use this bucket to host a website or redirect requests. Learn more

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps

Create Amplify app

S3 static website hosting
Enabled

Hosting type
Bucket hosting

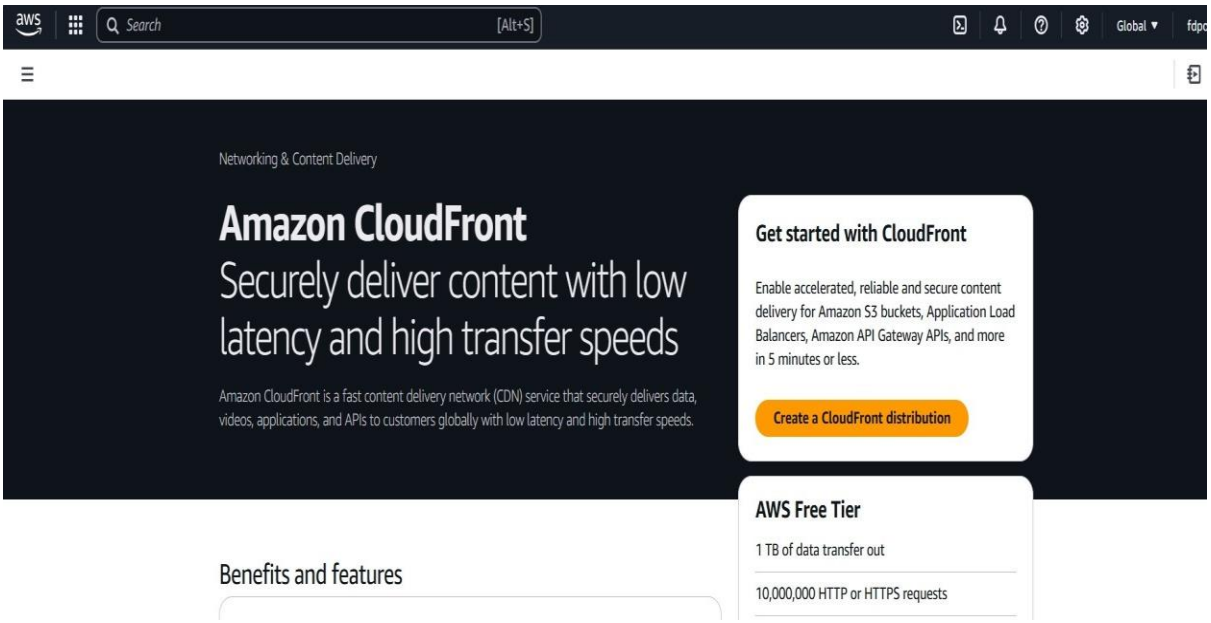
Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more
http://buchetforcloudfront.s3-website.eu-north-1.amazonaws.com

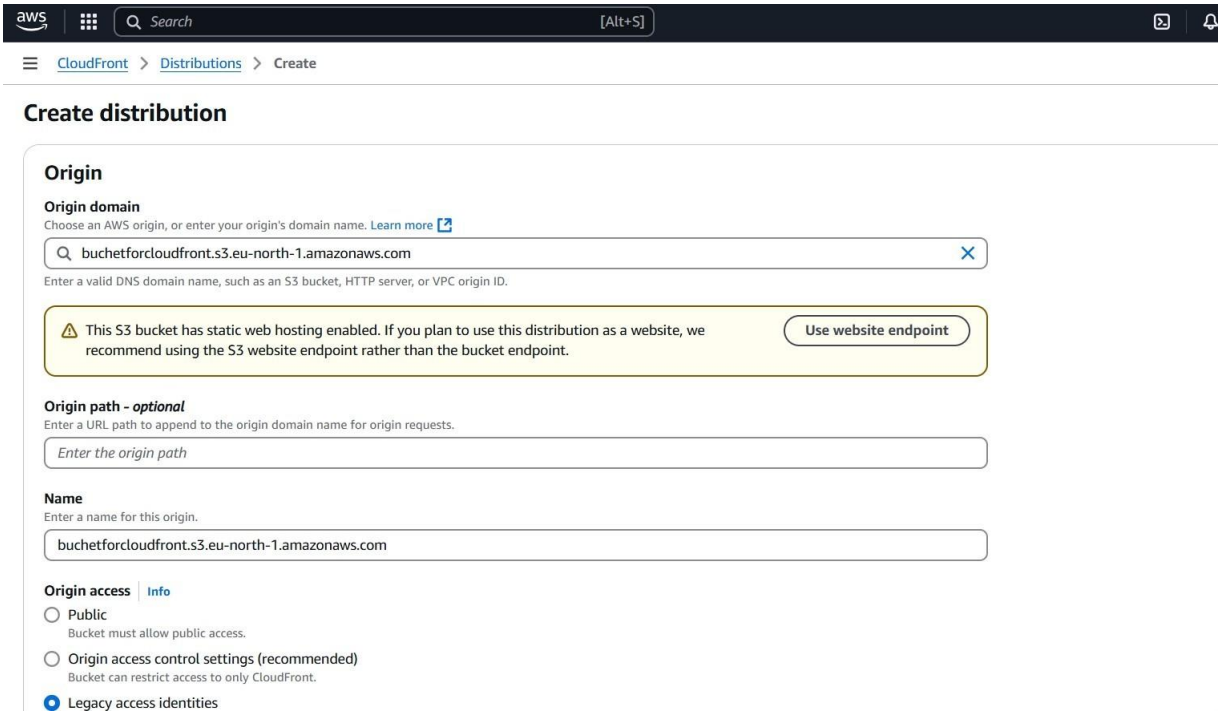
Not secure | buchetforcloudfront.s3-website.eu-north-1.amazonaws.com

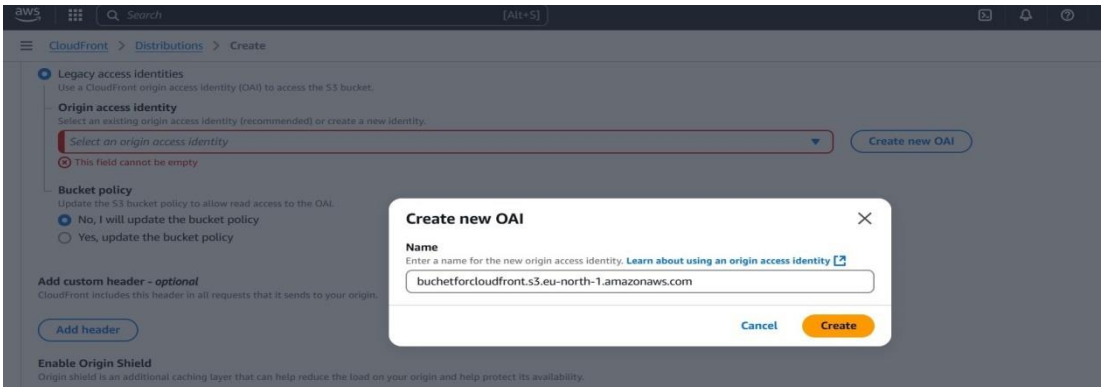
403 Forbidden

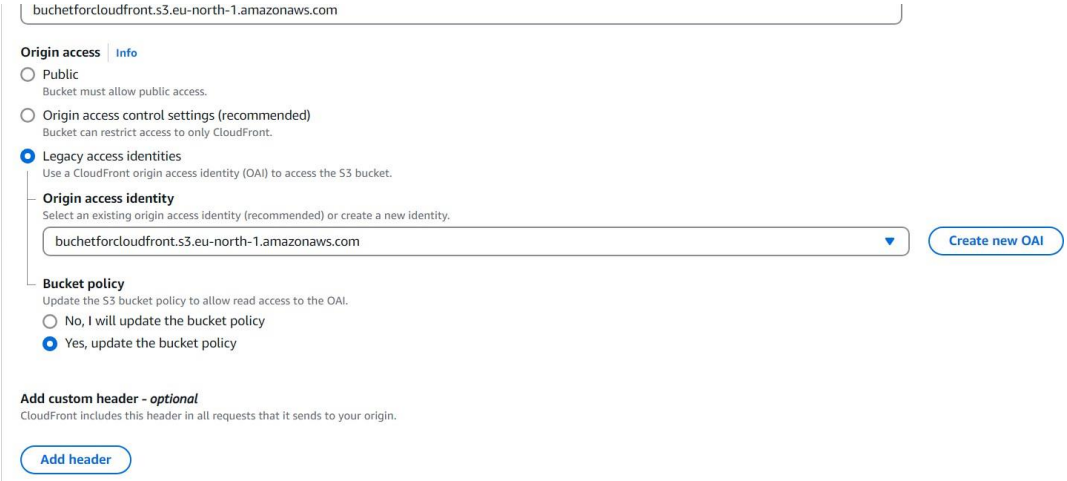
- Code: AccessDenied
- Message: Access Denied
- RequestId: YAF7TX6H40F421D2
- HostId: Irkdb7hf7HGG4X26oginZ7YKa1+YaRcxnsC5sqqVmyfsrYiuWoKLPmCM9jb9JRyW3Fb019ORyvWQ/P592gPrCw==

CLOUD FRONT


aws
Search
[Alt+S]
Global
Networking & Content Delivery
Amazon CloudFront
Securely deliver content with low latency and high transfer speeds
Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.
Get started with CloudFront
Enable accelerated, reliable and secure content delivery for Amazon S3 buckets, Application Load Balancers, Amazon API Gateway APIs, and more in 5 minutes or less.
Create a CloudFront distribution
AWS Free Tier
1 TB of data transfer out
10,000,000 HTTP or HTTPS requests
Benefits and features


aws
Search
[Alt+S]
CloudFront > Distributions > Create
Create distribution
Origin
Origin domain
Choose an AWS origin, or enter your origin's domain name. Learn more
buchetforcloudfront.s3.eu-north-1.amazonaws.com
Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.
This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint.
Use website endpoint
Origin path - optional
Enter a URL path to append to the origin domain name for origin requests.
Enter the origin path
Name
Enter a name for this origin.
buchetforcloudfront.s3.eu-north-1.amazonaws.com
Origin access
Info
Public
Bucket must allow public access.
Origin access control settings (recommended)
Bucket can restrict access to only CloudFront.
Legacy access identities


aws
Search
[Alt+S]
CloudFront > Distributions > Create
Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.
Origin access identity
Select an existing origin access identity (recommended) or create a new identity.
Select an origin access identity
Create new OAI
This field cannot be empty
Bucket policy
Update the S3 bucket policy to allow read access to the OAI.
No, I will update the bucket policy
Yes, update the bucket policy
Add custom header - optional
CloudFront includes this header in all requests that it sends to your origin.
Add header
Enable Origin Shield
Origin shield is an additional caching layer that can help reduce the load on your origin and help protect its availability.
Create new OAI
Name
Enter a name for the new origin access identity. Learn about using an origin access identity
buchetforcloudfront.s3.eu-north-1.amazonaws.com
Cancel
Create


buchetforcloudfront.s3.eu-north-1.amazonaws.com
Origin access
Info
Public
Bucket must allow public access.
Origin access control settings (recommended)
Bucket can restrict access to only CloudFront.
Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.
Origin access identity
Select an existing origin access identity (recommended) or create a new identity.
buchetforcloudfront.s3.eu-north-1.amazonaws.com
Create new OAI
Bucket policy
Update the S3 bucket policy to allow read access to the OAI.
No, I will update the bucket policy
Yes, update the bucket policy
Add custom header - optional
CloudFront includes this header in all requests that it sends to your origin.
Add header

Create distribution

Origin

Origin domain
Choose an AWS origin, or enter your origin's domain name. Learn more

buchetforcloudfront.s3.eu-north-1.amazonaws.com

Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.

⚠ This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint. [Use website endpoint]

Origin path - *optional*
Enter a URL path to append to the origin domain name for origin requests.

Enter the origin path

Name
Enter a name for this origin.

buchetforcloudfront.s3.eu-north-1.amazonaws.com

Origin access | Info

- ○ Public
  Bucket must allow public access.
- ○ Origin access control settings (recommended)
  Bucket can restrict access to only CloudFront.
- ◉ Legacy access identities
  Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access identity
Select an existing origin access identity (recommended) or create a new identity.

buchetforcloudfront.s3.eu-north-1.amazonaws.com [Create new OAI]

Bucket policy
Update the S3 bucket policy to allow read access to the OAI.

- ○ No, I will update the bucket policy
- ◉ Yes, update the bucket policy

Add custom header - *optional*
CloudFront includes this header in all requests that it sends to your origin.

[Add header]

Enable Origin Shield
Origin shield is an additional caching layer that can help reduce the load on your origin and help protect its availability.

- ◉ No
- ○ Yes

► Additional settings

Default cache behavior

Path pattern | Info

Default (*)

Compress objects automatically | Info

- ○ No
- ◉ Yes

Viewer

Viewer protocol policy

- ◉ HTTP and HTTPS
- ○ Redirect HTTP to HTTPS
- ○ HTTPS only

Allowed HTTP methods

- ◉ GET, HEAD
- ○ GET, HEAD, OPTIONS
- ○ GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

Restrict viewer access
If you restrict viewer access, viewers must use CloudFront signed URLs or signed cookies to access your content.

- ◉ No
- ○ Yes

Cache key and origin requests

We recommend using a cache policy and origin request policy to control the cache key and origin requests.

- ◉ Cache policy and origin request policy (recommended)
- ○ Legacy cache settings

Cache policy
Choose an existing cache policy or create a new one.

CachingOptimized — Recommended for S3
Policy with caching enabled. Supports Gzip and Brotli compression.

Create cache policy | View policy

Origin request policy - *optional*
Choose an existing origin request policy or create a new one.

Select origin policy

Create origin request policy

Response headers policy - *optional*
Choose an existing response headers policy or create a new one.

Select response headers

Create response headers policy

► Additional settings

Function associations - *optional* Info

Choose an edge function to associate with this cache behavior, and the CloudFront event that invokes the function.

	Function type	Function ARN / Name	Include body
Viewer request	No association		
Viewer response	No association		
Origin request	No association		
Origin response	No association		

Web Application Firewall (WAF) Info

- ○ Enable security protections
  Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.
- ○ Do not enable security protections
  Select this option if your application does not need security protections from AWS WAF.

▼ Included security protections

- Protect against the most common vulnerabilities found in web applications.
- Protect against malicious actors discovering application vulnerabilities.
- Block IP addresses from potential threats based on Amazon internal threat intelligence

Price estimate

► This AWS WAF configuration is estimated to cost $14 for 10 million requests/month

Settings

Anycast static IP list - optional | Info
Deliver traffic from a small set of IP addresses

There are no Anycast static IP lists available

Create an Anycast static IP list
There are no Anycast static IP lists available

Price class | Info
Choose the price class associated with the maximum price that you want to pay.

- ◉ Use all edge locations (best performance)
- ○ Use only North America and Europe
- ○ Use North America, Europe, Asia, Middle East, and Africa

Alternate domain name (CNAME) - *optional*
Add the custom domain names that you use in URLs for the files served by this distribution.

[Add item]

ⓘ To add a list of alternative domain names, use the bulk editor.

Custom SSL certificate - *optional*
Associate a certificate from AWS Certificate Manager. The certificate must be in the US East (N. Virginia) Region (us-east-1).

Choose certificate

and support for additional HTTP versions. HTTP/2.0 and HTTP/1.1 are supported by default.

- ☑ HTTP/2
- ☐ HTTP/3

Default root object - *optional*
The object (file name) to return when a viewer requests the root URL (/) instead of a specific object.

IPv6

- ○ Off
- ◉ On

Description - *optional*

Standard logging Info

Additional charges may apply. See Info for more details.

Log delivery
Get logs of viewer requests to CloudWatch, Amazon S3 or Firehose

- ◉ Off
- ○ On

[Cancel] [Create distribution]

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Default root object – *optional*
The object (file name) to return when a viewer requests the root URL (/) instead of a specific object.

/second.html

IPv6
- Off
- On

Description – *optional*

Standard logging Info
Additional charges may apply. See Info for more details.

Log delivery
Get logs of viewer requests to CloudWatch, Amazon S3 or Firehose
- Off
- On

Cancel | Create distribution

---

aws | Search [Alt+S] | Global | fdponcc

CloudFront > Distributions > E10OYDIV9P2N81

Introducing the CloudFront Security Dashboard
The new security tab is a unified place to configure, manage, and monitor security for your CloudFront distribution. The built-in dashboard gives you visibility into top security trends, allowed and blocked traffic, as well as visibility and controls for bots. CloudFront geographic restrictions are now part of the security dashboard.

Successfully created new distribution.
To get in-depth monitoring information for your distribution's internet traffic, create an Internet Monitor

Do you have 30 seconds to share your experience? Help us improve CloudFront.

E10OYDIV9P2N81

View metrics

General | Security | Origins | Behaviors | Error pages | Invalidations | Tags | Logging

Details

Distribution domain name
d21jmagi8xjsj.cloudfront.net

ARN
arn:aws:cloudfront::699475937818:distribution/E10OYDIV9P2N81

Last modified
Deploying

Settings

Edit

Description
-

Alternate domain names
-

Standard logging
Off

Activate Windows
Go to Settings to activate Windows.

---

aws | Search [Alt+S] | Europe (Stockholm) | fdponcc

Amazon S3 > Buckets > buchetforcloudfront

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access

Copy

```
{
  "Version": "2008-10-17",
  "Id": "PolicyForCloudFrontPrivateContent",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::cloudfront:user/CloudFront Origin Access Identity E14VD7XXB289B9"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::buchetforcloudfront/*"
    }
  ]
}
```

---

aws | Search [Alt+S] | Global | fdponcc

CloudFront > Distributions > E1WVDZ5WTBRLXD

E1WVDZ5WTBRLXD

View metrics

General | Security | Origins | Behaviors | Error pages | Invalidations | Tags | Logging

Distribution domain name copied

Distribution domain name
d3op513v5bf4mm.cloudfront.net

ARN
arn:aws:cloudfront::699475937818:distribution/E1WVDZ5WTBRLXD

Last modified
February 4, 2025 at 6:32:16 AM UTC

Settings

Edit

Description
-

Price class
Use only North America and Europe

Supported HTTP versions
HTTP/2, HTTP/1.1, HTTP/1.0

Alternate domain names
-

Standard logging
Off

Cookie logging
Off

Default root object
-

Continuous deployment Info

6csesyll.pdf | 172.16.0.2:2280/cportal/ip/user_l | CloudFront | Global | 403 Forbidden | ERROR: The request could not be

https://d21jmagi8xjsj.cloudfront.net

403 ERROR

The request could not be satisfied.

Bad request. We can't connect to the server for this app or website at this time. There might be too much traffic or a configuration error. Try again later, or contact the app or website owner.
If you provide content to customers through CloudFront, you can find steps to troubleshoot and help prevent this error by reviewing the CloudFront documentation.

Generated by cloudfront (CloudFront)
Request ID: dtkeV29VjY119bnroZ0lnhUBX8E1vnHxoxBfF9UFMLcMsKXs1wbK3A==